



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

(An Autonomous Institution Affiliated to Anna University, Chennai)

SATHYAMANGALAM - 638 401

Module Test - I - Apr - 2026

II Semester

22PH202 & ELECTROMAGNETISM AND MODERN PHYSICS

1	(i)	A student studies electric field lines around isolated positive and negative charges and observes distinct patterns in diagrams. The patterns differ significantly when multiple charges are introduced close to each other. The student attempts to relate these patterns to force interactions. The environment is ideal with no external disturbances. Integrate the observed field line patterns with the concept of electric force and infer how these visual representations indicate interaction behavior. (3 Marks - [Ap/C, 2])
2	(ii)	An engineering team must choose between representing electric interactions using force vectors or electric field concepts for system design. The system involves multiple charges and requires scalability. The team evaluates which approach provides better clarity and efficiency. Assess which representation is more effective and justify the reasoning based on system requirements. (2 Marks - [Ap/C, 1])
3	(i)	A timing circuit in an electronic device depends on predictable charging and discharging of a capacitor. Engineers must ensure consistent timing intervals despite environmental changes. The capacitor interacts with resistive elements to control signal delays. Material properties influence overall timing stability. Integrate the role of capacitance and material properties to infer how timing consistency is achieved and maintained in such circuits. (3 Marks - [An/C, 2])
4	(ii)	An electronics company selects capacitors for consumer devices requiring reliability and safety. The choice must consider dielectric strength, durability, and cost. Performance under varying voltages is critical. Engineers must balance technical and practical constraints. Assess how capacitor selection influences reliability and justify the professional considerations involved. (2 Marks - [U/C, 1])
5	(i)	A computational model is being developed to simulate electric field distribution for multiple charges. The system must calculate field intensity at various points in space. The charges are fixed and their positions are known. The model must follow a logical sequence to ensure accurate results. Formulate a logical procedure to compute electric field intensity at any point and justify the sequence of steps involved. (3 Marks - [An/C, 1])

6	(ii)	A system initially uses discrete charge calculations to determine electric field but plans to extend to continuous charge distributions. The complexity increases as the number of charges grows significantly. The designers must evaluate computational efficiency. Analyze the trade-off between discrete summation and continuous distribution methods and infer which approach is more efficient for large systems. (2 Marks - [Ap/C, 1])
7	(i)	A student assembles a circuit requiring a specific equivalent capacitance using multiple available capacitors. The available units vary in capacitance values. The student must logically combine them to achieve the desired outcome. Incorrect arrangement leads to mismatch in expected results. Construct a logical procedure to achieve the required equivalent capacitance using series and parallel combinations while ensuring correctness of arrangement. (3 Marks - [Ap/C, 2])
8	(ii)	A power system incorporates capacitors for energy buffering but faces limitations in size and heat dissipation. Increasing capacitance improves performance but adds bulk. Designers must evaluate efficiency versus scalability. Trade-offs are necessary for optimal system performance. Evaluate the trade-offs involved in increasing capacitance and infer how system efficiency is impacted. (2 Marks - [U/C, 1])
9	(i)	A charged spherical shell produces an electric field in surrounding space. A physicist models the flux through a closed surface enclosing the shell and attempts to relate it to the enclosed charge using integral expressions. They aim to verify whether the flux remains constant irrespective of surface radius. The situation requires linking surface integral formulation with charge distribution. Formulate the relationship between electric flux and enclosed charge and analyze how changing the Gaussian surface radius affects the computed flux. (5 Marks - [E/C, 2])
10	(i)	A research team observes unexpected electric field readings inside a supposedly shielded conducting chamber. The chamber is grounded, but sensors indicate non-zero field values at certain points. The chamber walls are made of conducting material but have minor imperfections and joints. External charged objects are placed near the chamber during testing. The team must identify the cause of shielding failure. Diagnose the possible reasons for the presence of electric field inside the chamber and infer how imperfections in conductor properties or grounding influence electrostatic equilibrium conditions. (5 Marks - [Ap/C, 2])
11	(i)	A laboratory setup measures electric flux through a tilted flat surface placed in a uniform electric field. The recorded flux values fluctuate unexpectedly when the surface orientation is changed slightly. The researchers suspect errors in interpreting angle dependence and projection of area vectors. They must diagnose whether the issue arises from incorrect mathematical formulation or conceptual misunderstanding of flux directionality. Analyze the inconsistency in measured flux values and infer the underlying conceptual or computational error related to surface orientation and field interaction. (5 Marks - [Ap/C, 2])

12	(i)	A dipole consisting of charges $+q$ and $-q$ separated by a small distance is placed in a uniform electric field. The system is free to rotate about its center. The angle between the dipole axis and the electric field varies during motion. Engineers are studying how torque depends on orientation for micro-actuator applications. They must model rotational behavior based on electric parameters. Formulate the relationship governing torque on the dipole and determine how angular variation influences the rotational tendency, integrating the concept of dipole moment into the analysis. (5 Marks - [An/C, 2])
13	(i)	An electrostatic sensor system is being designed in a research lab where point charges are arranged along a straight axis to generate a controlled electric environment. The design team observes that varying the spacing between charges significantly alters the work required to move a test charge between two points. They also note that maintaining a uniform electric field is critical for accurate sensing. The system must minimize energy consumption while ensuring measurable potential differences. The engineer must decide the optimal configuration using relationships between electric field and potential. Determine an optimal configuration strategy that minimizes energy expenditure while maintaining measurable potential difference, and justify how electric field–potential relationships guide this design decision. (5 Marks - [Ap/C, 2])
14	(i)	A copper wire of uniform cross-section is connected to a battery in a laboratory setup where the length of the wire is doubled while keeping the applied potential difference constant; simultaneously, temperature rises slightly due to prolonged current flow, altering charge carrier mobility; experimental readings show a reduction in current despite unchanged voltage; microscopic observations indicate variations in drift velocity; the system is analyzed to understand how geometry and temperature influence conduction; additional data suggests resistivity variation with temperature; the circuit is otherwise stable and free from external disturbances. Determine how the current changes quantitatively and infer the combined impact of length and temperature variation on drift velocity and resistance; integrate the relationships between resistivity, geometry, and microscopic charge motion to justify the observed reduction in current under constant voltage conditions. (5 Marks - [Ap/C, 2])
15	(i)	A high-voltage experimental setup exhibits unexpected variations in electric potential across a region where equipotential surfaces were assumed to be evenly spaced. Measurements reveal that the potential gradient is inconsistent, and charged particles deviate from predicted paths. The system includes both discrete charges and a continuous charge distribution. Engineers suspect an error in modeling the potential field. The inconsistency affects the calculated electrical energy stored in the system. Analyze the possible causes of the observed inconsistencies in potential gradient and particle motion, and infer how modeling errors in charge distribution affect system energy interpretation. (5 Marks - [An/C, 2])

16	(i)	An engineer designs a long-distance power transmission system using wires of varying materials where conductivity differs significantly; increasing wire thickness improves current capacity but raises cost and weight; environmental temperature variations affect resistivity across regions; the system must maintain efficient energy transfer with minimal power loss; alternative materials offer higher conductivity but lower mechanical durability; trade-offs between efficiency, cost, and reliability must be evaluated before final implementation. Evaluate the trade-offs involved in selecting conductor material and dimensions; prioritize design decisions that optimize power efficiency while balancing cost and environmental constraints, and justify the final selection using conductivity and resistance principles. (5 Marks - [Ap/C, 2])
17	(i)	A science communicator explains magnetism using a crowd analogy where each person represents a small magnetic region. Initially, people face random directions, representing an unmagnetized state. When music plays, people align in one direction, and when the music stops, the alignment disappears. This analogy is used to explain the behavior of magnetic materials. Map each element of the analogy to actual physical concepts such as domains, magnetization, and external field. Also, explain what causes the loss of alignment when the external influence is removed. (3 Marks - [Ap/C, 1])
18	(ii)	A physics textbook states that magnetic fields are produced only when electric charges move through a conductor. A researcher questions this by pointing out that some materials exhibit magnetism even without visible current flow. Identify the hidden assumption in the textbook statement and explain how atomic-level motion provides a more complete explanation of magnetism. (2 Marks - [U/C, 1])
19	(i)	An embedded systems engineer is designing a circuit that requires a specific equivalent capacitance using multiple capacitors of known values. The available components include capacitors of varying ratings, and the final design must meet strict space and performance constraints. The engineer must logically determine the arrangement strategy before implementation. Formulate the logical sequence required to achieve the desired equivalent capacitance using the available components while ensuring design constraints are satisfied. (3 Marks - [An/C, 2])
20	(ii)	A robotics system requires stable voltage supply during sudden load variations. Engineers consider adding capacitors to mitigate voltage drops but must decide specifications under limited design space and cost constraints. Recommend a strategic approach to selecting capacitor specifications to ensure voltage stability under the given constraints. (2 Marks - [Ap/C, 1])

21	(i)	During a classroom discussion, a student argues that since isolated electric charges exist and produce electric fields, isolated magnetic charges (monopoles) should also exist. The student supports this by explaining that even when a charged object is divided, charge still exists independently. Another student responds by cutting a bar magnet into smaller pieces and observing that each piece still has both north and south poles. Explain why the first argument appears logically correct based on electrostatics. Then, using experimental and theoretical reasoning, explain why isolated magnetic poles are not observed. (3 Marks - [Ap/P, 2])
22	(ii)	An engineer claims that magnetic fields can exist independently in space even without current or motion. Another engineer disagrees, stating that magnetic fields always originate from charge motion or intrinsic properties of particles. Evaluate the correctness of the first claim using principles of magnetic field generation and justify your answer. (2 Marks - [E/C, 1])
23	(i)	A communication circuit uses capacitors to filter noise and stabilize voltage signals in a data transmission system. Engineers integrate capacitors at multiple stages, observing that certain placements improve signal clarity while others cause unintended delays. The behavior is linked to charging and discharging cycles interacting with signal frequency. Integrate the concepts of charging and discharging processes with signal behavior to infer how capacitor placement affects system performance in the given circuit. (3 Marks - [Ap/C, 2])
24	(ii)	A consumer electronics company must choose between using capacitors or batteries for a wearable device that requires quick bursts of energy but minimal long-term storage. The decision impacts cost, reliability, and user experience. Assess the suitability of capacitors for this application and justify the decision considering professional design priorities. (2 Marks - [U/C, 1])
25	(i)	A “black box” system shows that magnetic flux density increases linearly with magnetic field intensity. When the internal medium is changed, the output doubles for the same input. The internal structure is unknown. The goal is to determine the governing relationship. Derive the relationship between B and H, identify the hidden parameter, and predict the output when $H = 50 \text{ A/m}$ in the second medium. (5 Marks - [Ap/P, 2])
26	(i)	A research facility is developing an electromagnetic launcher that relies on an inductor to store energy before release. The system currently operates with fixed inductance and current, but engineers are considering modifications to improve performance. Due to thermal constraints, increasing current is risky, while increasing inductance affects system size and cost. The design team must determine the most effective parameter to modify. Using the relationship between energy, inductance, and current, determine which parameter has a greater influence on stored energy and justify the most effective modification strategy under the given constraints. $U = \frac{1}{2} L I^2$ (5 Marks - [Ap/C, 2])

27	(i)	An engineer is designing a magnetic shielding system for a portable device. Material A has high permeability but is heavy and expensive, while Material B is lightweight with moderate permeability. The system must reduce magnetic field strength by at least 80%. The engineer must balance performance with portability. Analyze the trade-off between permeability and portability, propose the best design approach, and justify your answer using the relationship between magnetic flux density, magnetic field intensity, and permeability. (5 Marks - [An/P, 2])
28	(i)	In a transformer system, two coils are placed close to each other such that a changing current in the primary coil produces a varying magnetic field that links with the secondary coil. A student observes that voltage is induced in the secondary coil only when the current in the primary changes over time, and no voltage appears when the current remains constant. This principle is widely used in power transmission systems. Develop the mathematical expression for the induced emf in the secondary coil, explain its relation to Lenz's law, and justify why a constant current produces no induced emf. (5 Marks - [An/C, 2])
29	(i)	A startup is optimizing a magnetic rail launcher using a current-carrying rod. Two designs vary magnetic field strength and structural reinforcement. The system must produce at least 6 N force while avoiding elastic deformation. Trade-offs exist between electromagnetic force and mechanical stability. The goal is to choose the most viable design under constraints. : Using ($F = BIL$), compare how variations in both magnetic field and rod length influence force generation in each design, assess how structural reinforcement alters allowable operating limits, and justify which design (or modification) best satisfies both performance and durability requirements. (5 Marks - [Ap/C, 2])
30	(i)	A circuit with a semiconductor device operates under increasing voltage. Initially, current increases proportionally, but at higher voltages, current rises rapidly. The device temperature increases due to internal heating. The surrounding environment remains unchanged. Engineers suspect deviation from ideal electrical behavior. Diagnose the cause of deviation from Ohm's law in the system and infer how material properties and operating conditions influence resistance behavior. Justify your reasoning based on underlying electrical principles. (5 Marks - [Ap/P, 2])
31	(i)	A system with a current-carrying wire is modified by increasing current and distance. Additionally, a second identical wire is introduced. The problem explores how magnetic field scales and how multiple sources interact. Superposition becomes critical. : Evaluate how simultaneous changes in current and distance affect the magnetic field of each wire, determine the resultant field using superposition, and argue which parameter change has a greater impact on field strength. (5 Marks - [Ap/P, 2])

32	(i)	A cylindrical copper wire carries steady current under a uniform electric field. The conductor has fixed cross-sectional area and known charge carrier density. Engineers observe that electron drift velocity changes with applied field strength. Temperature is maintained nearly constant, ensuring stable material properties. The system is analyzed to relate microscopic motion to observable current. Analyze the relationship between current density and drift velocity, and determine how microscopic charge movement governs macroscopic current under the given conditions. Integrate material parameters into your reasoning and justify consistency with established electrical principles. (5 Marks - [An/C, 2])
33	(i)	A circuit designer observes that introducing a time-varying electric field between capacitor plates results in a measurable magnetic effect, even without conduction current. The system is analyzed under high-frequency conditions. The designer seeks to understand the underlying principle. Interpret how a time-varying electric field contributes to magnetic field generation and justify the concept introduced to resolve this observation. (3 Marks - [E/F, 2])
34	(ii)	A physicist studies electromagnetic interactions and observes that electric and magnetic fields appear interdependent in wave propagation. The system operates in free space with no charges present. Infer the relationship between electric and magnetic fields in free space and justify their role in wave propagation. (2 Marks - [E/P, 1])
35	(i)	A student observes that objects appear displaced when viewed through a glass slab. The slab has parallel faces and uniform thickness. The displacement seems consistent regardless of viewing angle. Determine the reason for apparent displacement and relate it to optical properties of the slab. Infer the governing principle. (3 Marks - [E/M, 2])
36	(ii)	A person with blurred distant vision uses corrective lenses to improve clarity. The eye lens fails to focus parallel rays properly on the retina. Identify the defect and determine the corrective measure required to restore proper vision. (2 Marks - [E/C, 1])
37	(i)	A coil is placed in a region where the magnetic field changes over time due to varying current in a nearby circuit. The coil experiences induced voltage, and the direction of induced current changes depending on the variation. The system operates without energy loss. Analyze the relationship between changing magnetic fields and induced electromotive force, and infer the principle governing direction of induced current. (3 Marks - [An/C, 2])

38	(ii)	A theoretical model integrates various electromagnetic laws into a unified framework to explain field interactions. The model successfully predicts wave propagation and field coupling. Analyze how multiple electromagnetic laws combine to form a unified theory and infer its physical significance. (2 Marks - [An/F, 1])
39	(i)	A fiber optic cable transmits signals using light pulses through a glass core. The surrounding cladding has a lower refractive index. Engineers observe that signals remain confined within the core even when the cable bends slightly. Analyze the mechanism that enables confinement of light within the core and infer the condition required for efficient transmission. (3 Marks - [An/P, 2])
40	(ii)	A ray of light travels from air into water at a specific angle. The refracted ray bends toward the normal. Infer the relative refractive indices and determine the relationship governing the bending behavior. (2 Marks - [An/P, 1])
41	(i)	An engineering team is designing a high-voltage capacitor system for energy storage in a pulsed power application. They observe that increasing the electric field strength improves storage but also raises insulation concerns. The dielectric material properties vary under high field conditions. The team needs to evaluate how energy is stored and distributed within the electric field. They also aim to optimize storage without breakdown. Analyze how the energy storage capability of the capacitor depends on the electric field and dielectric properties, and determine how design choices influence safe and efficient energy storage under the given conditions. (5 Marks - [E/P, 3])
42	(i)	A laboratory setup uses a monochromatic light source directed toward two narrow slits separated by a small distance, and the resulting pattern is projected on a distant screen. The experimenter observes alternating bright and dark fringes with uniform spacing. The wavelength of light and slit separation are known, and the distance to the screen is adjustable. Slight misalignment of slits changes the fringe clarity. Environmental vibrations also affect visibility. Analyze how fringe spacing and visibility are influenced by system parameters and determine the conditions required to maintain stable and distinct interference patterns in this setup. (5 Marks - [An/C, 3])
43	(i)	A wireless power transmission system uses oscillating currents to generate electromagnetic waves. Engineers notice that energy transfer efficiency decreases with distance. They analyze the role of electromagnetic fields and how energy propagates through space. The system operates in open air without guided structures. They must evaluate how energy flow is governed. Determine how electromagnetic energy propagates in the system and justify how the direction and magnitude of energy transfer are influenced by field interactions and spatial factors. (5 Marks - [E/C, 3])

44	(i)	An optical coating is applied on a glass surface to reduce reflection using a thin transparent film. The thickness of the film is comparable to the wavelength of visible light. Light incident on the surface undergoes multiple reflections within the film. Depending on wavelength and thickness, certain colors appear enhanced while others are suppressed. Evaluate how thin film thickness and refractive index determine the interference outcome and recommend conditions to minimize reflection for a specific wavelength. (5 Marks - [E/C, 3])
45	(i)	A physicist derives equations from Maxwell's laws to model wave propagation in free space. By combining curl equations, a second-order differential equation emerges. The solution predicts wave-like behavior traveling at a constant speed equal to the speed of light. Formulate how Maxwell's equations lead to the electromagnetic wave equation and infer the significance of the derived wave speed. (5 Marks - [E/P, 3])
46	(i)	A research lab is developing a medical laser system where atoms are excited using external energy sources. The system must maintain a higher population of atoms in excited states compared to ground states, and a mechanism is needed to sustain coherent photon emission. Determine the conditions required to achieve sustained laser action in the system and justify how atomic transitions contribute to coherent output generation. (5 Marks - [E/F, 3])
47	(i)	A research lab studies an oscillating electric field generated between capacitor plates where the field varies sinusoidally with time. Nearby, a magnetic field is detected forming perpendicular loops. The system is placed in vacuum, and energy transfer is observed without material contact. The scientists aim to understand how these fields sustain wave propagation. Analyze how time-varying electric and magnetic fields interact to produce a self-sustaining electromagnetic wave and determine the directional relationship between the fields and propagation. (5 Marks - [An/P, 3])
48	(i)	An industrial automation system uses polarized light sensors to detect stress patterns in transparent materials. During testing, different polarization states such as circular and elliptical are observed depending on phase differences introduced in the system. Analyze how different polarization states emerge in the system and infer how phase differences influence the final polarization form in practical applications. (5 Marks - [E/C, 3])